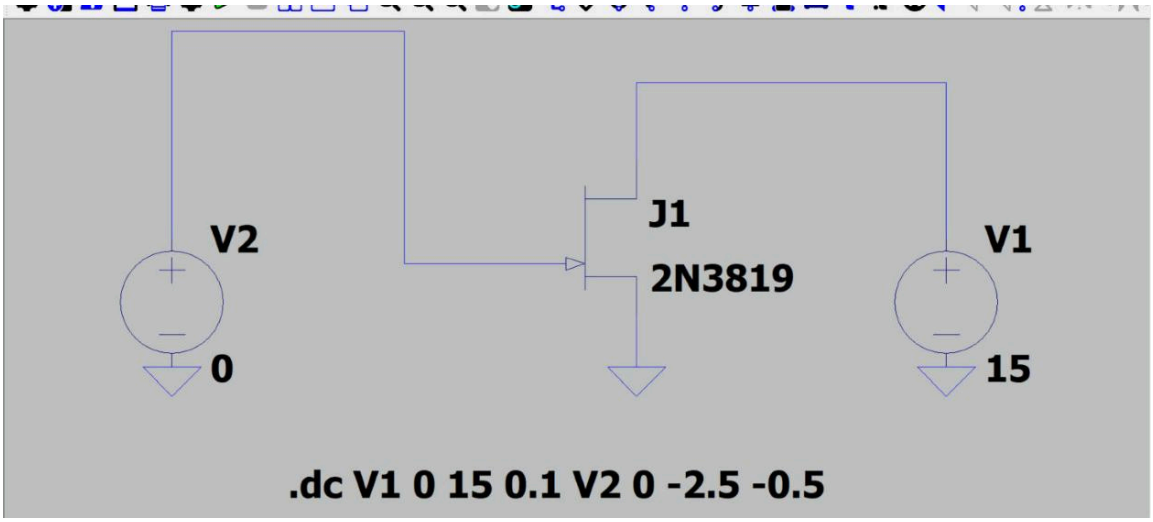
Project Part B — Q6(b): Output Characteristics of a JFET

Electronic Product Design & Reverse Engineering (26EC106PCB) · B.Tech ECE, Section B · Group 50
Jayesh Singh, Ayush Kumar, Shikha Saroj, Himanshi Kaushik, Anant Kumar

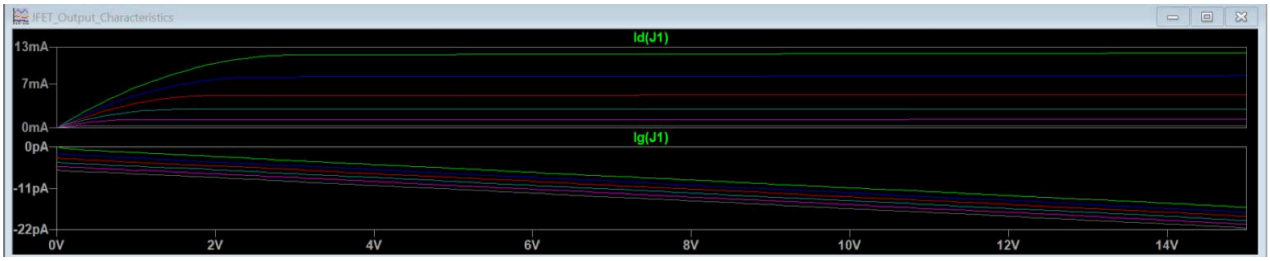
Problem statement. Simulate, observe and analyse the output characteristics of a JFET, infer its behaviour, and explain: (i) JFET is a voltage-controlled current source, (ii) JFET has high input impedance.

1. Simulation setup

Item	Value
Device	2N3819 n-channel JFET (LTspice standard model library)
Tool	LTspice (Analog Devices), DC sweep analysis
Circuit	Common-source: source grounded; V1 = drain supply (V_{DS}); V2 = gate supply (V_{GS})
Run 1: output characteristics	`.dc V1 0 15 0.1 V2 0 -2.5 -0.5`, i.e. V_{DS} 0 to 15 V in 0.1 V steps for V_{GS} = 0, –0.5, –1, –1.5, –2, –2.5 V (906 points)
Run 2: transfer characteristic	V_{DS} = 10 V, `.dc V2 -3.2 0 0.01`



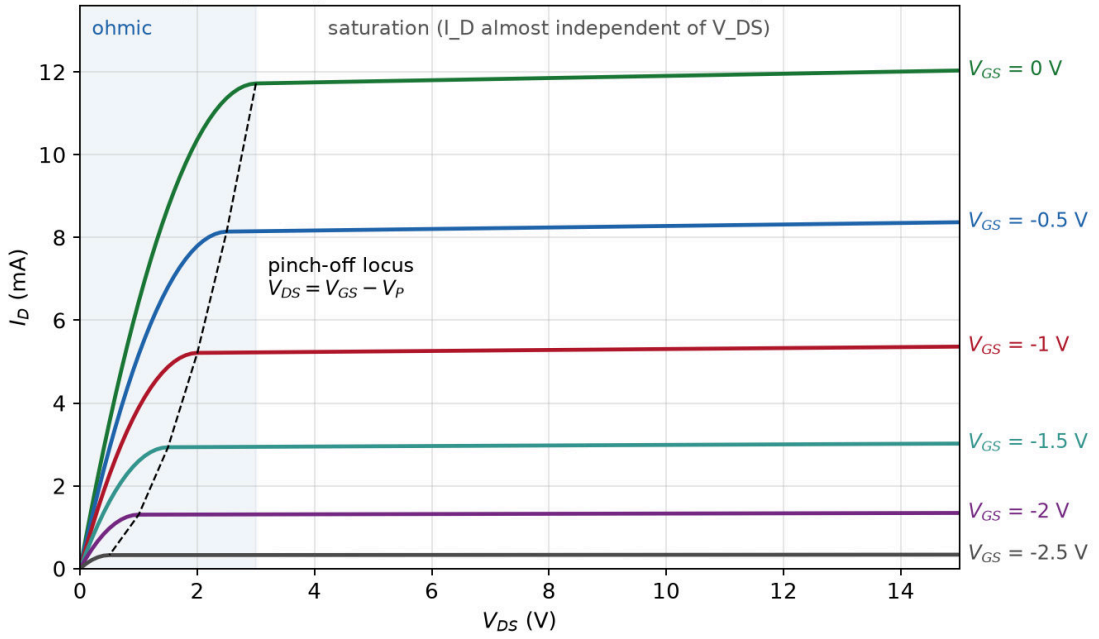
LTspice schematic used for Run 1.



LTspice waveform window: I_D (top) and I_G (bottom) for all six V_{GS} steps.

2. Output characteristics (I_D vs V_{DS})

Fig. 1 — Output characteristics of 2N3819 (LTspice data)



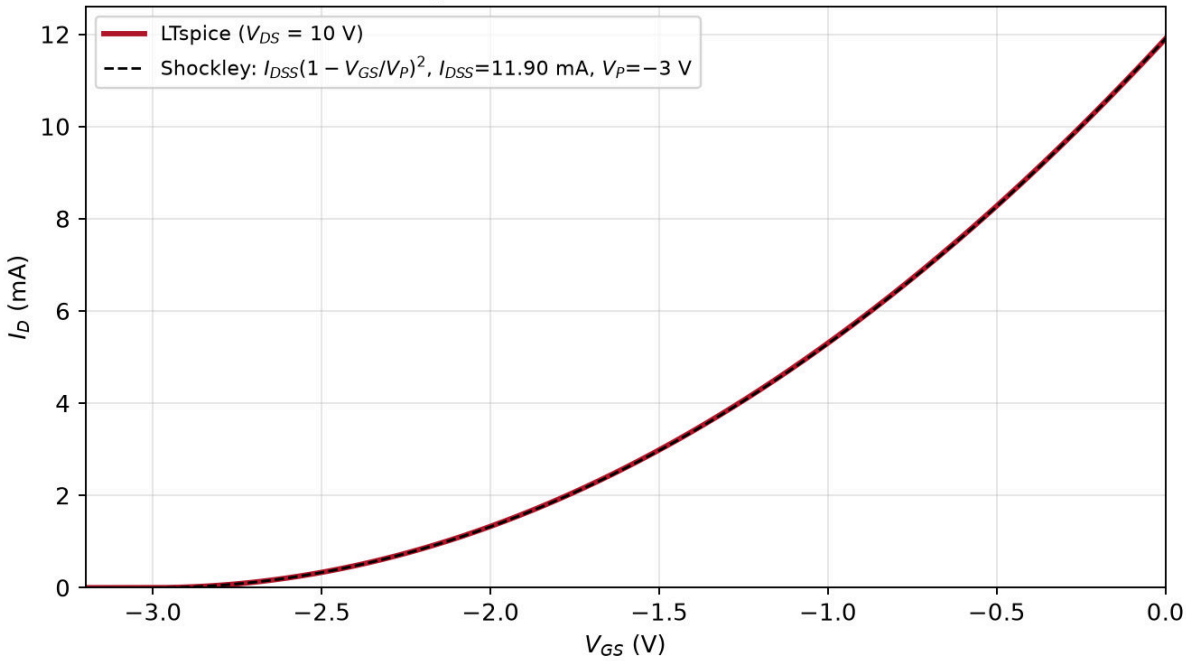
- **Ohmic region** (small V_{DS}): I_D rises almost linearly; the channel acts as a resistor whose value is set by V_{GS}.
- **Pinch-off:** each curve bends at $V_{DS} = V_{GS} - V_P$ (dashed line), with $V_P \approx -3$ V.
- **Saturation region:** curves are almost flat; I_D is fixed by V_{GS}, not by V_{DS}.
- **Cut-off:** for $V_{GS} \leq V_P \approx -3$ V, $I_D \approx 0$ (Run 2: 0.013 mA at –2.9 V; 0 at –3.0 V).

V_{GS} (V)	I_D at V_{DS} = 5 V (mA)	I_D at V_{DS} = 15 V (mA)	Change
0	11.77	12.03	2.2 %
-0.5	8.19	8.37	2.2 %
-1	5.25	5.36	2.2 %
-1.5	2.96	3.02	2.2 %
-2	1.32	1.34	2.2 %
-2.5	0.33	0.34	2.2 %

Tripling V_{DS} (5 → 15 V) changes I_D by only **≈ 2.2 %**, while moving V_{GS} from 0 to –2.5 V cuts I_D about **36 times**. All values are read directly from the LTspice output data (Id(J1)).

3. Transfer characteristic (I_D vs V_{GS})

Fig. 2 — Transfer characteristic



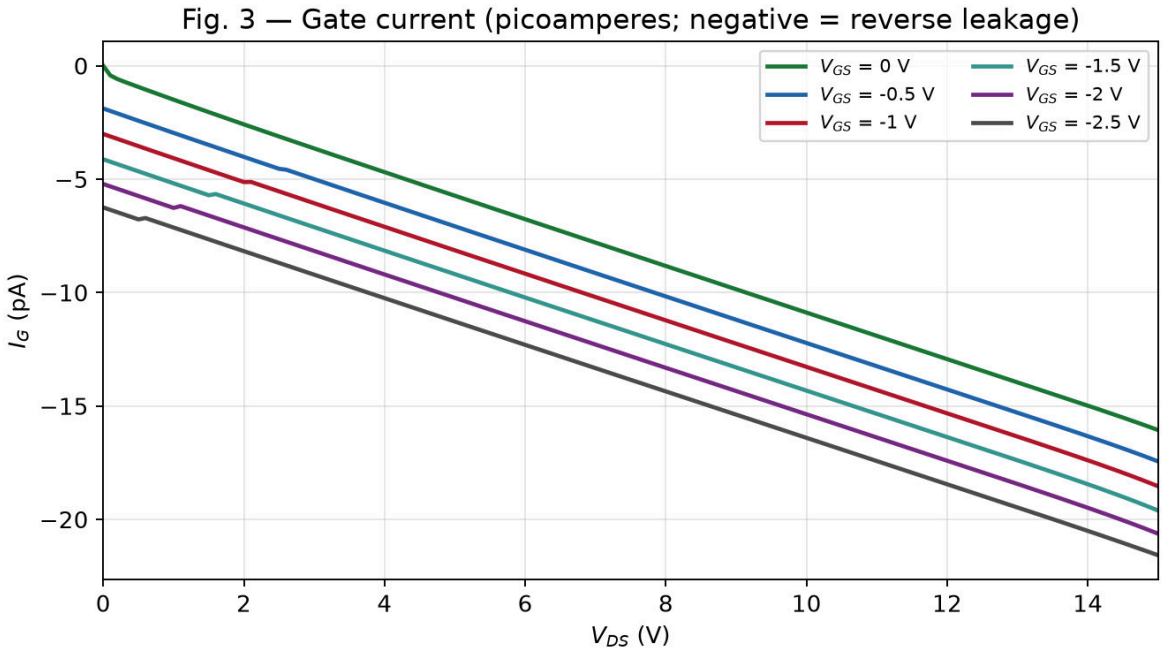
The curve is a parabola following **Shockley's equation** $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$, with I_{DSS} = 11.90 mA (at V_{DS} = 10 V) and V_P = –3 V. Check at V_{GS} = –1.5 V: 11.90 × (0.5)² = **2.98 mA**; the simulation gives **2.99 mA**.

Transconductance near V_{GS} = 0: $g_m = \Delta I_D/\Delta V_{GS}$ ≈ **7.9 mS** (theory $2I_{DSS}/|V_P|$ ≈ 7.9 mS). One volt at the gate controls milliamperes at the drain.

4. Inference (i): JFET is a voltage-controlled current source

- **Current source.** In saturation I_D stays almost constant (≈ 2.2 % change) while V_{DS} triples, just as an ideal current source delivers a fixed current whatever the voltage across it.
- **Voltage-controlled.** Which constant current flows is set only by V_{GS} (0 V → 11.8 mA, –1 V → 5.2 mA, –2 V → 1.3 mA), and the gate draws practically no current (Section 5), so the control is by voltage, not by current.
- Small-signal form: $i_d = g_m \cdot v_{gs}$, the defining relation of a VCCS. (Contrast: a BJT is current-controlled, $I_C = \beta \cdot I_B$.)

5. Inference (ii): JFET has high input impedance



- The gate–channel p–n junction is **reverse-biased** ($V_{GS} \le 0$), so only a tiny leakage current flows into the gate.
- Largest simulated gate current: $|I_G| \approx$ **21.6 pA** ($V_{GS} = -2.5$ V, $V_{DS} = 15$ V). It grows with V_{DS} because the gate–drain junction is reverse-biased by $V_{DS} - V_{GS}$.
- Input resistance $R_{in} \approx |V_{GS}|/|I_G| = 2.5$ V / 21.6 pA ≈ **116 GΩ (~10^{11} Ω)**.
- Datasheet cross-check (2N3819): $I_{GSS} \le 2$ nA at $V_{GS} = -15$ V, giving $R_{in} \ge 7.5$ GΩ.
- I_D is in milliamperes, I_G in picoamperes (about 10^9 times smaller), so the JFET does not load the signal source.

6. Conclusions

1. The output characteristics show ohmic, saturation and cut-off regions, with $I_{DSS} \approx 11.8$–11.9 mA and $V_P \approx -3$ V.
2. In saturation I_D is nearly independent of V_{DS} and set only by V_{GS}, so the JFET is a **voltage-controlled current source**.
3. The gate current is in picoamperes, so the **input resistance is ~10^{11} Ω**.
4. The transfer curve agrees with Shockley's equation.

Applications: JFET-input op-amps (TL071/TL081), electret-microphone pre-amplifiers, high-impedance test probes and pH meters, constant-current sources.

Simulation files (.asc schematics, .raw data, logs) are in the `ltspice/` folder of this repository; opening an .asc file in LTspice and pressing Run reproduces every graph.